Pharmacy Sales & Inventory Analysis — 2025

Michael Sillyman · Data Analyst portfolio project · Tools: SQL (SQLite), LibreOffice Calc · Simulated dataset for a US retail pharmacy

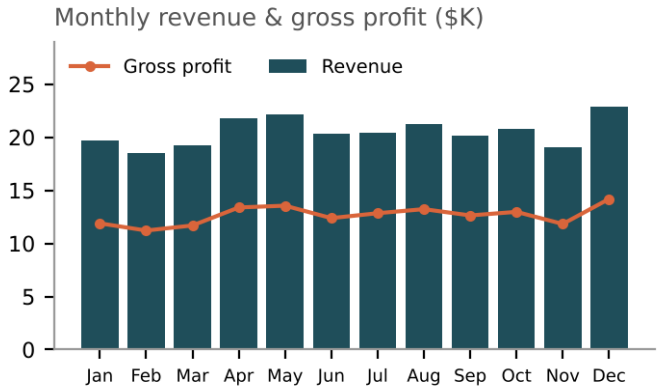
Business question

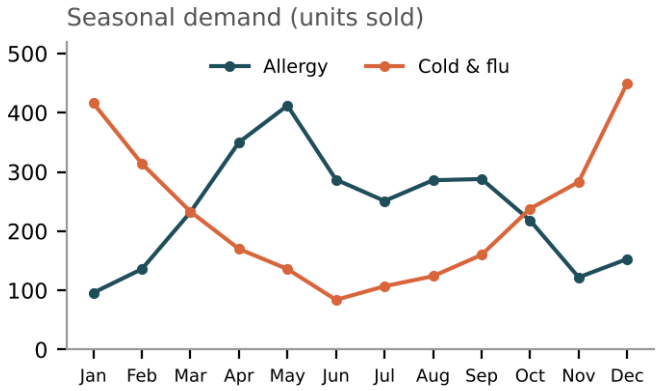
Which products and categories drive revenue and profit, how does demand change through the year, and where is stock either running short or tying up cash? The goal is a data-backed restocking plan.

Data & approach

A relational database of 3 tables (products, inventory, sales) holding 15,084 transactions across 30 products (OTC and generic prescriptions) in 11 categories. Ran data-quality checks (invalid quantities, prices, orphan records, below-cost pricing: all passed), wrote 7 SQL queries for sales, margin, seasonality and stock cover, then rebuilt the core summaries in LibreOffice Calc with SUMIFS/COUNTIFS formulas and charts so the results update when the data changes.

Revenue	Gross profit	Gross margin	Transactions	Avg. sale
$246.1K	**$151.8K**	**61.7%**	**15,084**	**$16.32**





Key findings

- **Pain Relief is the largest category** at 20.3% of revenue ($49.8K), and acetaminophen is the top seller at $23.2K. Allergy follows at 17.7%, while cardiovascular prescriptions carry the highest margin (69.2%).
- **Demand is strongly seasonal.** Allergy sales peak in May (412 units), 4.3× the Jan low. Cold & flu sales peak in Dec (450 units), 5.4× the Jun low.
- **Dec was the strongest month** at $22.9K, driven by holiday traffic and flu season. Gross margin rose from 60.9% in H1 to 62.4% in H2 after the July price review.
- **6 products are below their reorder level** at year end, 2 of them cold & flu items in peak flu season. The cold & flu liquid has 14 units against a reorder level of 30. Two chronic-care prescriptions, Metformin and Lisinopril, are also short.
- **$1,518 is tied up in 5 slow-moving items**, mostly medical devices; the blood pressure monitor has 2,208 days of cover at current sales. Sunscreen is also on the list, but its demand is seasonal and will pick up in spring.
- **82% of transactions are cashless** (card and mobile wallet).

Recommendations

1. **Plan stock by season:** build allergy inventory in February–March ahead of the April–May peak, and cold & flu inventory from October.
2. **Restock the 6 alert items now**, starting with the cold & flu products and acetaminophen, then the chronic-care prescriptions.
3. **Pause device reorders** and promote blood pressure monitors and pulse oximeters to patients on blood pressure medication; hold sunscreen for spring.
4. **Review reorder levels every quarter** using the days-of-cover query, so levels track real demand.